

The logic of arguments - Set 3 - Solutions

Specification Arg1-Arg4.

Answer key

Question	Answer	Topic
1	A	The logic of arguments

Worked solutions

Question 1 LGC-15 · The logic of arguments · spec Arg2

Let P : 'the integer n is divisible by 4' and Q : 'the integer n^2 is divisible by 8'. Which one of the following is true?

Answer: A

If $4 \mid n$ then $16 \mid n^2$, so $8 \mid n^2$: P is sufficient. Conversely, if $8 \mid n^2$ then n is even, say $n = 2m$, and $8 \mid 4m^2$ forces m to be even, so $4 \mid n$: P is also necessary. (Tempting wrong answer: 'sufficient but not necessary' - but no counterexample exists.)

Full method - how and why

A genuine 'necessary **and** sufficient' case, dressed to look like the usual 'sufficient only'. You must prove both directions rather than assume a counterexample exists.

1 Sufficiency: if $4 \mid n$, write $n = 4k$. Then $n^2 = 16k^2$, which is divisible by 8. **True.**

why: 16 is a multiple of 8, so divisibility by 8 follows immediately.

2 Necessity: suppose $8 \mid n^2$. Then n^2 is even, so n is even - write $n = 2m$.

why: An odd number squared is odd, so n cannot be odd.

3 Then $n^2 = 4m^2$, and $8 \mid 4m^2$ forces $2 \mid m^2$, so m is even, say $m = 2j$.

why: Divide the divisibility through by the common factor 4.

4 So $n = 4j$, i.e. $4 \mid n$. Both directions hold: necessary **and** sufficient.

why: No counterexample exists, which is what distinguishes this from the usual pattern.

Where it goes wrong

- Answering 'sufficient but not necessary' out of habit, without actually hunting for a counterexample. Test $n = 2$: $n^2 = 4$, not divisible by 8 - so it is not a counterexample; $n = 6$: 36, not divisible by 8. The hunt keeps failing, which is the clue.
- Assuming $8 \mid n^2 \Rightarrow 8 \mid n$ - false ($n = 4$ gives $n^2 = 16$).

Takeaway Before declaring a condition 'not necessary', actually look for a counterexample; repeated failure to find one is a signal to prove the converse instead.